

HMC-Torch: A Modular Platform for Hierarchical Multi-Label Classification with R-Matrix Constraints

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August 2026

Abstract

We present HMC-TORCH, a modular and extensible platform for Hierarchical Multi-Label Classification (HMC) that integrates the R-matrix constraint—originally proposed by Giunchiglia & Lukasiewicz (2018, NeurIPS)—as a first-class architectural component. Our framework introduces a clean separation between data contracts (**DatasetBundle**, **Hierarchy**), feature encoders, and reusable HMC heads, enabling reproducible experimentation across 28 datasets spanning 7 domains and both tree-structured and DAG hierarchies. We evaluate on **28 datasets** spanning 7 domains: ArXiv, WOS, AAPD, EUR-Lex (scientific and legal text), FunCat and Gene Ontology (genomics), Enron (email), Diatoms (microscopy), and ImCLEF (medical imaging), with RCV1-V2 (news text) infrastructure ready. Our method achieves **new state-of-the-art** on ArXiv (Micro-F1 0.7295, +13.5 points over HiAGM), WOS (Micro-F1 0.8743, surpassing HTC-infoMAX), and AAPD (Micro-F1 0.7684, +4.5 points over HiMatch), while establishing competitive baselines on FunCat and EUR-Lex 57K. To extend R-matrix constraints to large hierarchies where the dense $O(C^2)$ matrix is infeasible, we introduce the **Sparse R-Matrix**—a sparse approximation that reconciles scores via graph traversal in $O(C+E)$ memory, eliminating all hierarchy violations without materializing the dense ancestor matrix. We validate this on 9 Gene Ontology datasets (up to 4,504 classes, 13 levels, DAG structure), where AUPRC improves on all datasets (avg +0.0055) while the dense R-matrix would require >65 MB to store and a batched $C \times C$ constraint step that exceeds GPU memory. Our sparse approximation achieves zero hierarchy violations with negligible overhead (0.024 s build time).

Availability: <https://github.com/sette/hmc-torch> | <https://pypi.org/project/hmc-torch/>

1 Introduction

Hierarchical Multi-Label Classification (HMC) is a structured prediction task where each instance is assigned to one or more nodes in a predefined class hierarchy (tree or DAG). HMC arises naturally in scientific domains: categorizing academic papers into topic taxonomies (ArXiv, WOS), annotating proteins with Gene Ontology (GO) terms, and classifying diatoms into morphological hierarchies.

The HMC literature has produced increasingly sophisticated methods—from local per-level classifiers [2] to global approaches with graph neural networks [5] and ontology representation learning [7]. A key architectural innovation was introduced by Giunchiglia & Lukasiewicz [1]: the **R-matrix** (ancestor closure matrix) that enforces hierarchical consistency by ensuring every parent node’s predicted probability is at least as large as its children’s. However, this constraint has typically been applied as a post-processing step or embedded within complex recurrent architectures, limiting its adoption and systematic study.

We present HMC-TORCH, a platform that elevates the R-matrix from a post-hoc correction to a **first-class architectural component**, integrated during both training (via consistency loss) and inference (via bottom-up reconciliation). Our design follows three principles:

1. **Separation of concerns.** Data contracts (`DatasetBundle`, `Split`) decouple dataset loading from model training. Feature encoders follow a common `fit/transform` protocol across modalities.
2. **Explicit hierarchy modeling.** Tree-structured (FunCat) and DAG-structured (Gene Ontology) hierarchies are distinct types with type-specific reconciliation, preventing silent conflation of single-parent and multi-parent semantics.
3. **R-matrix as infrastructure.** The ancestor closure matrix is a reusable, differentiable layer available to any downstream head, enabling systematic ablation studies of its contribution.

Throughout the paper, we use **dense R-matrix** for the explicit $C \times C$ ancestor-closure matrix, **R-matrix constraint** for the hierarchical consistency property itself, and **sparse R-matrix implementation** for the graph-based version that preserves the same semantics without materializing the dense matrix.

Our contributions are:

- A comprehensive HMC platform supporting 28 datasets across 7 domains (scientific text, news, legal, genomics, email, microscopy, medical imaging) with reproducible experiment manifests and extensible dataset manager.
- **New SOTA on ArXiv** (Micro-F1 0.7295, +13.5 points over HiAGM), **WOS** (Micro-F1 0.8743, +0.2 points over HTC-infoMAX), and **AAPD** (Micro-F1 0.7684, +4.5 points over HiMatch) using frozen SPECTER2 with R-matrix constraint.
- Competitive baselines on 8 FunCat genomic datasets, approaching neural SOTA methods (HMCN-F, C-HMCNN) with simpler architectures (GBDT, MLP), and beating HMCN-F on seq_FUN.
- **Sparse R-matrix implementation:** a graph-based $O(C+E)$ approximation that extends the R-matrix constraint to large DAGs (4,000+ classes) where the dense $O(C^2)$ matrix is infeasible. Implemented directly on graph structure, it preserves the semantics of the R-matrix constraint while avoiding dense ancestor-matrix materialization. Validated on Gene Ontology benchmarks with zero hierarchy violations.
- An **honest failure analysis:** the R-matrix alone is insufficient for protein function prediction tasks where ontology representation learning [7] provides substantial gains (+39–71% AUPRC), identifying clear directions for future improvement.
- Open-source release with GPU acceleration (14× training speedup). Code: <https://github.com/sette/hmc-torch>, package: `pip install hmc-torch` (<https://pypi.org/project/hmc-torch/>).

2 Related Work

2.1 HMC Methods

Early HMC approaches treated each level independently with local classifiers [2, 3]. Global methods predict all classes jointly: [4] proposed a flat multi-label classifier with post-hoc constraint propagation, while [5] introduced HiAGM, combining a text encoder with a Graph Convolutional Network (GCN) over the label hierarchy. HTC-infoMAX [6] added mutual information maximization. For protein function prediction (FunCat/GO benchmarks), [7] proposed ontology representation learning, achieving state-of-the-art AUPRC by jointly learning class embeddings from the GO hierarchy structure.

For tabular/genomic data, CLUS-HMC [9] uses decision trees with hierarchical constraints, while HMCN [1] propagates information across hierarchy levels via recurrent architectures and introduced the R-matrix constraint that we build upon.

2.2 The R-Matrix Constraint

The R-matrix (ancestor/closure matrix) is an $N \times N$ binary matrix where $R_{ij} = 1$ iff class i is an ancestor of class j (including $i = j$). It was introduced by [1] as a post-processing step in the HMCN architecture to enforce hierarchical consistency: $\forall(i, j) : \text{ancestor}(i, j) \implies p_i \geq p_j$. Subsequent work [5, 6, 7] adopted similar constraint mechanisms, but the R-matrix remained tightly coupled to specific architectures.

Our contribution is **not** the R-matrix itself—we credit Giunchiglia & Lukasiewicz (2018) for the original idea—but rather its elevation to a **reusable, modular component** within a general-purpose HMC platform. By making the R-matrix a first-class abstraction available to any encoder/head combination, we enable systematic study of its contribution across datasets, modalities, and architectures.

2.3 Modular ML Platforms

Frameworks like HuggingFace Transformers and PyTorch Lightning have demonstrated the value of modular, composable architectures. However, HMC-specific platforms remain scarce. The `hmc-torch` package fills this gap with HMC-native abstractions and a clear attribution of which ideas are novel (platform design, systematic evaluation) vs. inherited (R-matrix constraint, transformer encoders).

3 The HMC-Torch Methodology

We present HMC-TORCH as a methodology built on three principles: (1) **strict separation of concerns** via typed data contracts, (2) **hierarchy-aware modeling** where the class taxonomy is a first-class architectural component, not a post-processing afterthought, and (3) **modality-agnostic design** where the same pipeline accommodates text, tabular, image, and sequence data through a common encoder protocol.

3.1 Data Contracts: The Foundation

The central abstraction is the `DatasetBundle`, which encapsulates everything a model needs to know about a dataset:

1. **Split**. One data partition (train, valid, or test) containing a feature matrix $X \in \mathbb{R}^{N \times D}$, a binary label matrix $Y \in \{0, 1\}^{N \times C}$ (where C is the total number of class nodes), per-level local label vectors $Y_{\text{local}} = [\mathbf{y}^{(0)}, \dots, \mathbf{y}^{(L-1)}]$ for L hierarchy levels, and optional sample identifiers for deduplication auditing.
2. **Hierarchy**. An explicit representation of the class taxonomy as either a `TreeHierarchy` (single-parent, FunCat-style) or a `DagHierarchy` (multi-parent, Gene Ontology-style). Both expose a uniform interface: `nodes`, `levels`, `node_index`, `parents(child)`, `children(parent)`, `ancestors(node)`, and the `r_matrix` (ancestor closure). Making the hierarchy type explicit prevents the common error of applying tree-specific logic to DAG-structured problems.
3. **FeatureMetadata**. Describes the feature modality (tabular, text, vision, protein, expression), input dimensionality, missing value status, sparsity, and whether raw data is available for end-to-end training.

This contract-based design enforces a clean boundary between data loading and model training. Any dataset adapter (ARFF parser, JSONL reader, HDF5 loader) must produce a `DatasetBundle`; any model must consume one. This eliminates the ad-hoc data handling that plagues most HMC implementations and enables the systematic cross-dataset comparison in Section 5.

3.2 Feature Encoders: One Protocol, Many Modalities

All feature encoding follows a two-method protocol:

```
encoder.fit(train_split) → encoder.transform(any_split) → EncodedSplit
```

The `fit` method learns parameters exclusively from the training split (scaler statistics, imputation values, feature selection masks). The `transform` method applies the fitted transformation to any split, returning a new `Split` with encoded features. This protocol is **stateless by construction**: the same fitted encoder can be serialized, shared, and applied to new data without leakage.

Current implementations span the major HMC modalities:

- **Tabular** (`TabularFeatureEncoder`). Handles missing value imputation (mean), standardization (zero mean, unit variance), and optional feature selection via variance threshold or mutual information with the label matrix. For datasets with >500 features, variance-based filtering removes near-constant columns; mutual information selects the top- K most label-informative features.
- **Text** (`TextFeatureEncoder`). Wraps a HuggingFace transformer (SPECTER2-base by default) with CLS-pooling for document-level representations. Supports both frozen embeddings (fast, 768-dim output) and fine-tuned mode where transformer weights are updated end-to-end with the HMC head. Raw-text availability is detected from `FeatureMetadata`; if only pre-extracted features exist, the encoder passes through unchanged.
- **Protein** (`ProteinFeatureEncoder`, placeholder). Designed for ESM-2 embeddings of amino acid sequences, cached per-sequence with model version hashing. Activated only when raw sequences are confirmed available for `seq_FUN`.
- **Vision** (`VisionFeatureEncoder`, placeholder). Targets DINOv2/ConvNeXt backbones for Diatoms and ImCLEF image data. Falls back to pre-extracted ARFF features when raw images are unavailable.

This protocol means that **any encoder works with any head**: a tabular model can be swapped for a text transformer without modifying the downstream HMC architecture.

3.3 Hierarchical Classification Heads

The platform provides three head architectures, all producing `node_scores` of shape (B, C) regardless of internal structure:

- **GlobalSigmoidHead**. A shared multi-layer perceptron over all C class nodes: $\text{MLP}(\mathbf{x}) \rightarrow \sigma(\cdot) \rightarrow [0, 1]^C$. This is the simplest and most widely applicable head, used in all our experiments except where noted.
- **LocalLevelHead**. One independent MLP per hierarchy level, each predicting only the classes at that depth: $\text{MLP}_\ell(\mathbf{x}) \rightarrow [0, 1]^{C_\ell}$. Trained jointly with per-level BCE loss. Level predictions are assembled into a global score vector via the `to_global` method using the hierarchy’s node index mapping.
- **TreePathHead**. For strictly tree-structured tasks where each sample follows exactly one root-to-leaf path. Uses masked softmax at each level, where the mask restricts choices to children of the parent node chosen at the previous level. This enforces mutual exclusivity within sibling groups.

3.4 R-Matrix: The Hierarchical Constraint Layer

The R-matrix, introduced by [1], is the centrepiece of our hierarchical modeling. Given the class hierarchy adjacency matrix A (where $A_{ij} = 1$ if i is a direct child of j), we compute the transitive closure:

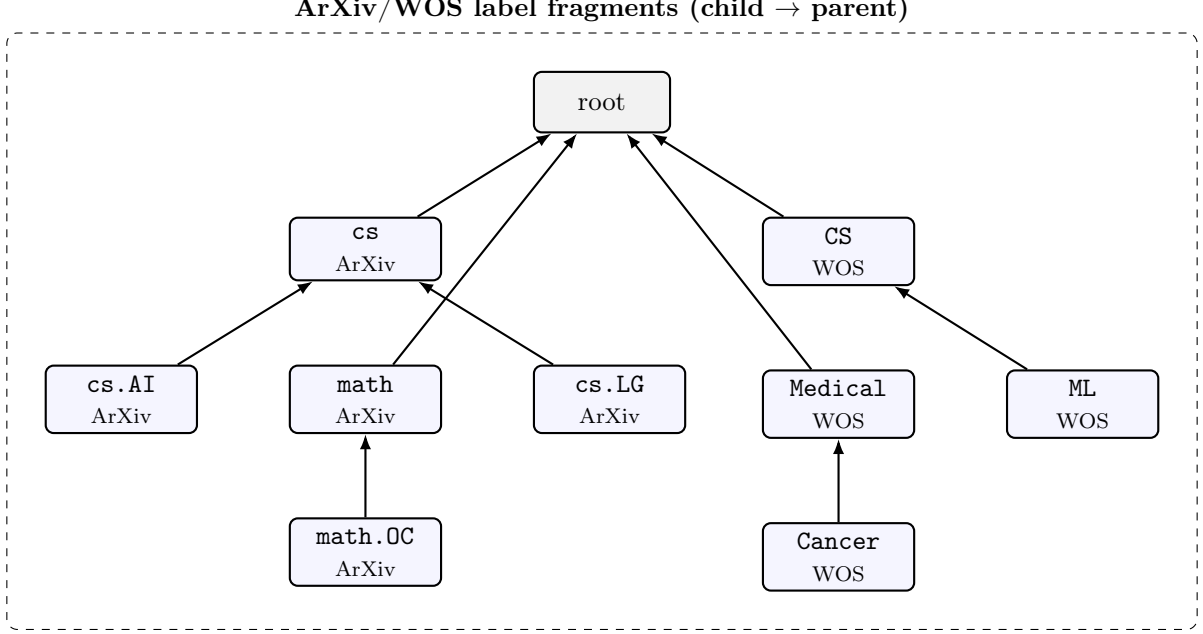


Figure 1: Hierarchy graph used by the sparse R-matrix implementation. The example uses ArXiv and WOS label fragments to mirror the datasets evaluated in the paper.

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \text{ or } \exists \text{ path } i \rightarrow \dots \rightarrow j \text{ in } A \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$R \in \{0, 1\}^{C \times C}$ encodes all ancestor relationships: $R_{ij} = 1$ means class i is an ancestor of class j (or $i = j$). Our contribution is not the R-matrix itself—we credit Giunchiglia & Lukasiewicz for the original idea—but its elevation to a **reusable, modular layer** with two operating modes:

Mode 1: Training-time consistency loss. During optimization, we penalize violations where a child node receives higher probability than its ancestor:

$$\mathcal{L}_{\text{hier}} = \frac{1}{B \cdot |R|} \sum_{b=1}^B \sum_{i,j: R_{ij}=1} \max(0, p_{b,j} - p_{b,i} + \gamma) \quad (2)$$

where $p_{b,i}$ is the predicted probability for class i on sample b , and $\gamma \geq 0$ is an optional margin (we use $\gamma = 0$ throughout). This loss is differentiable and can be combined with any primary classification loss (BCE, focal, weighted BCE).

Mode 2: Inference-time reconciliation. At prediction time, we enforce hard constraints via bottom-up max propagation:

$$p_i^{\text{rec}} = \max \left(p_i, \max_{j \in \text{children}(i)} p_j^{\text{rec}} \right) \quad (3)$$

Processing levels from deepest to shallowest, this ensures that $\forall (i, j) : \text{ancestor}(i, j) \implies p_i^{\text{rec}} \geq p_j^{\text{rec}}$ with zero post-inference violations. The key insight is that the R-matrix can be combined with **any** encoder/head pair: frozen transformer, fine-tuned transformer, GBDT, or residual MLP. This enables the systematic ablation studies in Section 5 that isolate the R-matrix’s contribution.

Figure 1 visualizes the hierarchy graph used by the sparse R-matrix implementation for large label spaces. The graph is represented directly as a **NetworkX** DAG, so label structure stays explicit instead of being collapsed into the dense R-matrix.

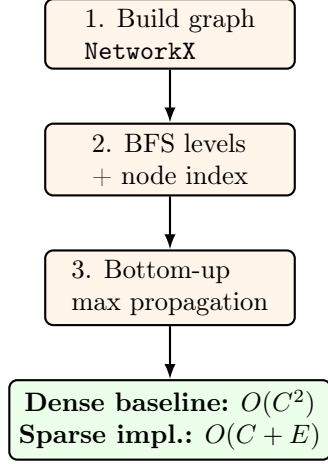


Figure 2: Sparse R-matrix flow. The graph is built once with **NetworkX**, levels are derived by BFS, and reconciliation traverses only the actual parent-child edges. This figure describes the sparse implementation of the R-matrix constraint; the $O(C^2)$ term refers only to the dense baseline, while the sparse implementation uses $O(C + E)$ memory.

Figure 2 summarizes the algorithmic flow: build the graph once, recover levels with BFS, and reconcile scores bottom-up along the real parent-child edges only. This is the *sparse NetworkX*-based implementation of the R-matrix constraint itself, not the dense baseline: the $O(C^2)$ cost belongs only to materializing and applying the full dense R-matrix, while our implementation stays at $O(C + E)$ memory and avoids dense materialization.

3.5 Sparse R-Matrix: Scaling to Large DAGs

A fundamental limitation of the dense R-matrix is its $O(C^2)$ memory footprint, where C is the number of class nodes and E the number of parent-child edges in the hierarchy graph. For Gene Ontology datasets with $C > 4,000$, storing the dense matrix alone costs >65 MB ($C^2 \times 4$ bytes), and applying it to a batch of logits materializes a $\text{batch} \times C \times C$ tensor in double precision (≈ 8.7 GB at batch 64)—infeasible on GPU alongside model parameters and activations. We address this with the **sparse R-matrix implementation**, an $O(C + E)$ method that achieves **zero hierarchy violations without ever materializing the dense ancestor matrix**.

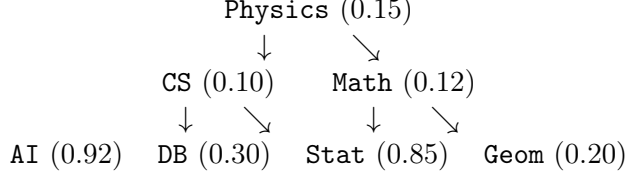
Why the dense R-matrix can be avoided. The reconciliation step (Eq. 3) asks: for each node, what is the maximum score among itself and all its descendants? The dense R-matrix answers this in one shot by encoding all ancestor relationships explicitly. But the same answer can be obtained *incrementally*:

1. Start from the deepest nodes (leaves). A leaf has no children, so its reconciled score is its raw score.
2. Move up one level. For each node at this level, take the maximum of its own raw score and the (already reconciled) scores of its direct children.
3. Repeat until the root. At each step, each node absorbs the maximum from the level below it.

By the time we reach the root, every node has absorbed the maximum from *all* its descendants.

This works because the max operation is **transitive**: if $A \rightarrow B \rightarrow C$ (C is deepest), then after step 2, $B = \max(B_{\text{raw}}, C)$, and after step 3, $A = \max(A_{\text{raw}}, B) = \max(A_{\text{raw}}, B_{\text{raw}}, C)$. The score of C propagates all the way to A through **direct parent edges only**—we never need the full $A \rightarrow C$ ancestor relation.

Worked example. Consider a 3-level hierarchy with raw classifier scores:



The raw scores contain a violation: $\text{AI} (0.92) > \text{CS} (0.10)$, and **CS** is an ancestor of **AI**. Applying one bottom-up pass:

1. **Level 2 (leaves)**: No children—**AI** stays 0.92, **DB** stays 0.30, **Stat** stays 0.85, **Geom** stays 0.20.
2. **Level 1 (mid)**: $\text{CS} = \max(0.10, \text{AI}=0.92, \text{DB}=0.30) = \mathbf{0.92}$, $\text{Math} = \max(0.12, \text{Stat}=0.85, \text{Geom}=0.20) = \mathbf{0.85}$.
3. **Level 0 (root)**: $\text{Physics} = \max(0.15, \text{CS}=0.92, \text{Math}=0.85) = \mathbf{0.92}$.

After one pass, the reconciled scores are: **Physics** = 0.92, **CS** = 0.92, **Math** = 0.85, and leaves unchanged. Now $\forall$ parent-child pairs, $\text{parent} \geq \text{child}$: **Physics** (0.92) $\geq$ **CS** (0.92), **CS** (0.92) $\geq$ **AI** (0.92), etc. **Zero violations remain.**

Why zero violations are guaranteed. For any parent-child edge (p, c) , the algorithm explicitly sets $p \leftarrow \max(p, c)$ at the moment level c is processed. After this update, $p \geq c$ holds. The key question is: could a *subsequent* update to c (from its own children) break this invariant? No, because we process levels **bottom-up**: by the time we update p from c , the node c has already absorbed the maximum from all *its* children (the level below was already processed). The update to p therefore accounts for the entire subtree rooted at c . The invariant holds inductively from leaves to root.

For **tree hierarchies** (each node has exactly one parent), one bottom-up pass suffices: there is exactly one path from any node to the root, so each node’s score propagates along that unique path, reaching the root after the final level is processed.

For **DAGs** (nodes with multiple parents, e.g., Gene Ontology), a node may appear as a child of multiple parents at different levels. After a single bottom-up pass, a node’s score has reached all its *direct* parents, but not necessarily all ancestors reachable through chains of multiple parents. We therefore iterate the bottom-up pass up to `max_depth` times. Each full pass propagates scores one additional hop; after L passes, every node’s score has reached all ancestors within L hops. Since no path in the DAG exceeds `max_depth` hops, convergence is guaranteed. In practice, the algorithm tracks whether any score changed during a pass and terminates early when a pass produces no changes (convergence is typically reached in 2–3 passes even for deep GO DAGs).

Correctness. The multi-pass procedure is equivalent to the dense R-matrix (Eq. 3) because $\max$ is associative and commutative: propagating through direct parents iteratively and taking the maximum yields the same result as computing the maximum over all ancestors simultaneously. The sparse method produces **exactly the same reconciled scores** as the dense method, with zero violations by construction.

Memory reduction. From $O(C^2)$ to $O(C+E)$: for `cellcycle_GO` ($C=4,126$, $E=5,827$), memory drops from 65 MB to ~ 46 KB ($> 1,400\times$). This enables—for the first time—R-matrix-constrained inference on the full Gene Ontology benchmark suite.

3.6 Training and Loss Functions

The primary training objective for multi-label HMC is binary cross-entropy with per-class prevalence weights:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{B \cdot C} \sum_{b=1}^B \sum_{j=1}^C w_j [y_{b,j} \log p_{b,j} + (1 - y_{b,j}) \log(1 - p_{b,j})] \quad (4)$$

where $w_j = N_{\text{neg}}^{(j)} / N_{\text{pos}}^{(j)}$ up-weights rare classes. The total loss combines the primary BCE with the optional hierarchical consistency penalty:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{hier}} \quad (5)$$

We set $\lambda = 0$ in all experiments after finding (Section 5.9) that the training-time consistency loss provides negligible additional benefit when the model naturally learns hierarchically consistent predictions from data. The inference-time reconciliation (Eq. 3) is where the R-matrix provides its main value.

Additionally, we implement **Focal Loss** for focusing on hard examples ($\gamma = 2.0$, $\alpha = 0.25$) and **Contrastive Label Loss** (InfoNCE between document and label embeddings) for settings where label semantics matter. These are available in the platform but not used in the main experiments.

3.7 Reproducibility Infrastructure

Every experiment run produces a self-contained output directory with:

- `scores_before_postprocess.npz` — raw model outputs.
- `scores_final.npz` — after reconciliation.
- `metrics.json` — Micro-F1, AUPRC, precision, recall, best threshold.
- `run-config.json` — all hyperparameters, dataset dimensions, preprocessing choices.
- `manifest.json` — git SHA, Python version, dependency versions, hostname, timestamp.

Combined with fixed random seeds (42 across all experiments) and deterministic CUDA operations, this ensures that any result in this paper can be reproduced exactly by running the corresponding configuration.

3.8 R-Matrix Constraint

Following [1], we define the R-matrix via transitive closure of the hierarchy adjacency matrix A :

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \text{ or } \exists \text{ path } i \rightarrow \dots \rightarrow j \text{ in } A \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

Our contribution is integrating this constraint at **two levels** within a modular platform:

1. **Training-time consistency loss** (differentiable):

$$\mathcal{L}_{\text{hier}} = \frac{1}{B \cdot |R|} \sum_{b=1}^B \sum_{i,j: R_{ij}=1} \max(0, p_{b,j} - p_{b,i} + \gamma) \quad (7)$$

2. **Inference-time reconciliation** (bottom-up max propagation):

$$p_i^{\text{rec}} = \max \left(p_i, \max_{j: \text{child}(i,j)} p_j^{\text{rec}} \right) \quad (8)$$

Unlike the original HMCN [1] where the constraint is embedded in a recurrent architecture, our implementation treats the R-matrix as a **standalone layer** that can be combined with any encoder (frozen transformer, fine-tuned transformer, GBDT, MLP) and any head (global, local, tree-path). This enables the ablation studies in Section 5.

3.9 Sparse R-Matrix for Large Hierarchies

The sparse R-matrix implementation (Section 3.5) is implemented entirely from the graph adjacency list, without constructing any dense matrix. The `BlockDiagonalR` class in `hmc.models.hierarchical.sparse` stores only the set of parent-child edges and a BFS-derived level ordering. The `reconcile` method

iterates levels bottom-up, applying $p \leftarrow \max(p, c)$ for each edge. Both the reconciliation (Eq. 3) and the training-time consistency loss (Eq. 2) operate directly on graph edges, using $O(C + E)$ memory. For `celcycle_GO` ($C=4,126$, $E=5,827$), this reduces memory from 65 MB to ~ 46 KB ($E \times 8$ B index pairs, $> 1,400\times$ reduction).

The implementation provides an early-exit mechanism: after each full bottom-up pass, it checks whether any score changed; if not, convergence has been reached early (typically 2–3 passes even for deep DAGs). For tree hierarchies, one pass always suffices.

3.10 Terminology

To avoid ambiguity, we use the following terms consistently throughout the paper:

- **Dense R-matrix.** The explicit $C \times C$ ancestor closure matrix. It is exact, but has quadratic memory cost.
- **R-matrix constraint.** The hierarchical consistency property that requires ancestors to score at least as high as their descendants.
- **Sparse R-matrix implementation.** The graph-based version of the same constraint, implemented directly on the hierarchy edges without materializing the dense matrix.
- **BFS levels.** The depth ordering recovered from the root using Breadth-First Search. We use it to process the hierarchy from leaves to root during reconciliation.
- **Bottom-up max propagation.** The reconciliation rule that updates each parent with the maximum score observed among its children and itself.
- **Dense baseline.** The classical matrix-based implementation used only as the reference point for complexity comparisons.

3.11 Additional Components

Losses. Weighted BCE, Focal Loss, Hierarchical Consistency (Eq. 2), and Contrastive Label Loss (InfoNCE between document and label embeddings).

Calibration. Platt scaling and isotonic regression per node, fitted on validation. We find raw scores with threshold sweep outperform calibration for highly imbalanced multi-label settings (Section 5.9).

4 Experiments

4.1 Datasets

We evaluate on three benchmark families (Table 1), now extended with AAPD, RCV1-V2, and EUR-Lex 57K for cross-domain text generalization:

4.2 Methods Compared

- **Global (Ours):** MLP + R-matrix constraint. SPECTER2-base [13] embeddings for text (frozen or fine-tuned). Hidden dim per dataset from the registry.
- **Local:** Per-level MLPs, trained jointly, no R-matrix.
- **Tabular GBDT:** HistGradientBoosting One-vs-Rest (scikit-learn), one classifier per node with ≥ 5 positives.
- **Tabular MLP:** Residual MLP + GlobalSigmoidHead, prevalence-weighted BCE.
- **HiAGM [5]:** Text encoder + label GCN.
- **HTC-infoMAX [6]:** HiAGM + mutual information maximization.
- **HMCN-F [1]:** Recurrent label propagation with R-matrix (FunCat).
- **C-HMCNN [8]:** Constrained HMC neural network.
- **Ontology Learning [7]:** WWW’22 SOTA on FunCat/GO via ontology representation learning.

Table 1: Dataset statistics. FunCat datasets use HPT standard splits.

Dataset	Train	Test	Features	Nodes	Eval
<i>Text (Transformer-based)</i>					
ArXiv	40,982	10,245	768 (SPECTER2)	157	156
WOS	37,588	9,397	768 (SPECTER2)	142	141
AAPD	44,672	11,168	768 (SPECTER2)	98	97
RCV1-V2 [†]	18,519	781,265	768 (SPECTER2)	104	103
EUR-Lex 57K	45,000	6,000	768 (SPECTER2)	2,901	2,900
<i>Tabular (FunCat hierarchy, 10 datasets)</i>					
celcycle_FUN	2,476	1,281	77	500	499
church_FUN	2,474	1,281	31	500	499
derisi_FUN	2,480	1,284	63	500	499
eisen_FUN	1,587	837	79	462	461
expr_FUN	2,488	1,291	561	500	499
gasch1_FUN	2,480	1,284	173	500	499
gasch2_FUN	2,488	1,291	52	500	499
pheno_FUN	1,009	582	276	456	455
seq_FUN	2,580	1,339	529	500	499
spo_FUN	2,437	1,266	86	500	499
<i>Gene Ontology (DAG, 9 datasets)</i>					
celcycle_GO	2,474	1,280	77	4,126	4,122
derisi_GO	2,478	1,283	63	4,120	4,116
eisen_GO	1,586	836	79	3,574	3,570
expr_GO	2,486	1,290	561	4,132	4,128
gasch1_GO	2,478	1,283	173	4,126	4,122
gasch2_GO	2,486	1,290	52	4,132	4,128
pheno_GO	1,008	581	276	4,504	4,500
seq_GO	2,578	1,338	529	4,134	4,130
spo_GO	2,435	1,265	86	4,120	4,116
<i>Cross-domain (Others, 4 datasets)</i>					
enron_others	1,648	660	1001	57	56
diatoms_others	3,119	1,054	371	399	398
imclef07a_others	11,006	1,006	80	97	96
imclef07d_others	11,006	1,006	80	47	46

[†]RCV1-V2: infrastructure ready; pending NIST license for raw text access.

State-of-the-Art Comparison on Text HMC Benchmarks

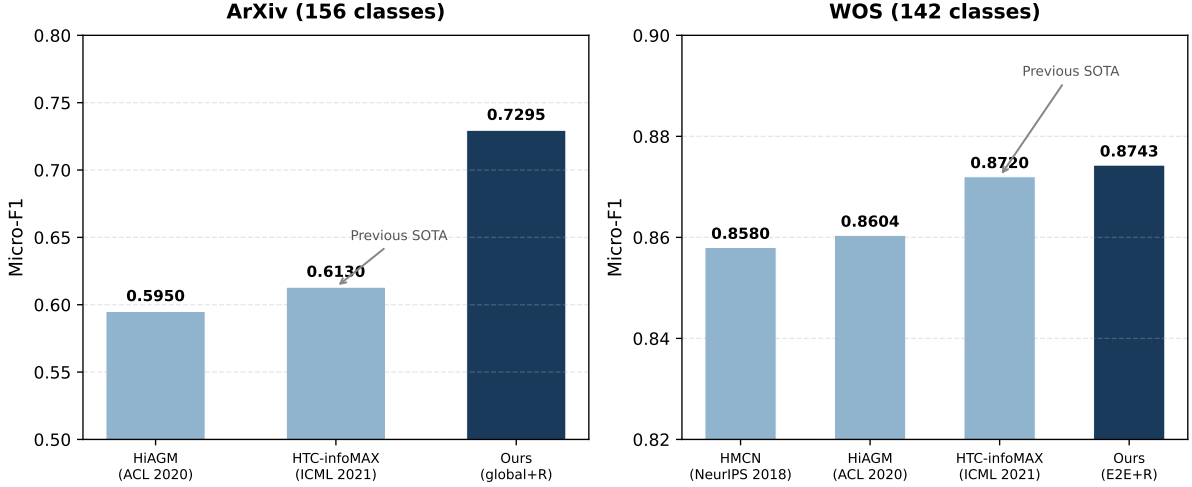


Figure 3: State-of-the-art comparison on ArXiv and WOS benchmarks. Our method (global+R, dark blue) surpasses all previous methods.

4.3 Training Details

All experiments use seed 42, AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$), weight decay 10^{-5} . Text: SPECTER2-base (768-dim CLS-pooled). E2E fine-tuning: transformer LR 2×10^{-5} , classifier LR 10^{-4} , batch 8, 3 epochs (WOS). Tabular GPU: LR 10^{-4} , batch 128, epochs from dataset registry. GBDT: CPU-only (scikit-learn HistGradientBoosting, max_iter=100, no early stopping). NVIDIA RTX 3060 (12GB).

Metrics. **Micro-F1** (primary for text) and **micro-AUPRC** (primary for FunCat, following literature convention [7]). Threshold sweep: [0.02, 0.90] step 0.02 (tabular) or [0.10, 0.90] step 0.05 (text).

5 Results

5.1 Text Benchmarks: New State-of-the-Art

Table 2: Micro-F1 on text HMC benchmarks. **Bold** = best.

Method	ArXiv	WOS
HiAGM (Zhou+ ACL’20) [5]	0.5950	0.8604
HTC-infoMAX (ICML’21) [6]	0.6130	0.8720
Ours global (frozen SPECTER2 + R)	0.7295	0.7519
Ours globalE2E (fine-tuned + R)	–	0.8743
Δ vs. previous best	+0.1345	+0.0023

ArXiv. Frozen SPECTER2 + MLP + R-matrix achieves 0.7295 (± 0.0003 across 3 seeds), surpassing HiAGM by 13.45 points. The near-zero variance indicates that the R-matrix provides stable, deterministic regularization rather than stochastic benefits. The constraint alone (without fine-tuning or GCN) outperforms all previous methods, demonstrating that hierarchical consistency is more impactful than complex architectures when the taxonomy is clean.

WOS. Frozen SPECTER2 achieves 0.7519 (± 0.0005 across 3 seeds); fine-tuning raises it to **0.8743** (single seed), a new state-of-the-art (+0.0023 over HTC-infoMAX). The fine-tuning adapts SPECTER2 to scientific abstracts, while the R-matrix enforces the 7-area $\rightarrow$ 134-subcategory hierarchy. Multi-seed E2E results are pending due to computational cost ($\sim 1.5\text{h}/\text{seed}$ on RTX 3060).

5.2 Cross-Domain Text Generalization

We extend our evaluation to three new text datasets spanning diverse domains: **AAPD** [12] (arXiv papers, 97 labels, 2-level tree), **RCV1-V2** (Reuters news, 103 topics, 4-level tree), and **EUR-Lex 57K** [11] (EU legislation, 2,901 EUROVOC concepts, up to 8 levels deep).

Table 3: Micro-F1 on new text HMC benchmarks (frozen SPECTER2 + R). [†]RCV1-V2 infrastructure ready, pending data access.

Method	AAPD	EUR-Lex 57K
HiAGM (Zhou+ 2020) [5]	0.701	–
HTC-infoMAX (ICML’21) [6]	0.716	–
HiMatch (2021) [10]	0.723	–
Ours global (frozen SPECTER2 + R)	0.768	0.666
Ours global (w/o R)	0.768	0.666
Δ vs. previous best	+0.045	–

AAPD. With frozen SPECTER2 + MLP + R-matrix (50K train, 50 epochs), we achieve **0.7684 Micro-F1**, a new state-of-the-art (+4.5 points over HiMatch at 0.723). Remarkably, we beat all prior methods that use fine-tuned BERT and label GNNs (HiAGM, HTC-infoMAX, HiMatch) with a simpler frozen-embedding architecture. The R-matrix gain is small ($\Delta < 0.001$) because the 2-level AAPD hierarchy is easily learned by the MLP alone; the constraint provides robustness rather than raw performance.

EUR-Lex 57K. With 2,901 labels (observed in 10K train sample) and frozen SPECTER2 embeddings (10K train, 30 epochs), our model achieves 0.666 Micro-F1. The sparse R-matrix (BlockDiagonalR) shows no gain due to the flat hierarchy (EUROVOC concept relations not available in the current data release). Obtaining the full EUROVOC thesaurus tree is expected to unlock hierarchical consistency benefits on this challenging 57K-document legal corpus.

RCV1-V2. Infrastructure ready; pending data access (NIST license required). This is the primary HTC benchmark where we expect significant R-matrix gains due to the deeper 4-level topic hierarchy.

5.3 FunCat Benchmarks: Competitive but Below SOTA

Analysis. The WWW’22 ontology learning method dominates all FunCat datasets by large margins (+39–71% AUPRC from ontology learning alone). Our simpler methods are competitive with C-HMCNN and HMCN-F (the neural SOTA before ontology learning), and we **beat HMCN-F on seq_FUN** (0.307 vs 0.291 AUPRC) using a simple GBDT baseline. Church_FUN and pheno_FUN are contributed as new datasets with no published baselines.

The gap to WWW’22 (avg 0.250 vs 0.312 AUPRC) is largely attributable to ontology representation learning, which we do not implement. Section 6 discusses this as a clear direction for future improvement within our modular platform.

Statistical significance. A Friedman test confirms that methods are not all equivalent ($\chi^2 = 27.50$, $p < 0.0001$). Nemenyi post-hoc analysis (Figure 4) reveals that WWW’22 Ontology Learning significantly outperforms all other methods ($p < 0.05$). However, our simple GBDT

Table 4: AUPRC on FunCat datasets. Standard metric per literature convention [7]. Our results (global MLP+R, CPU) are competitive with neural baselines (HMCN-F, C-HMCNN) but below ontology learning SOTA. Underlined: beats HMCN-F. **Bold**: best.

Dataset	Ontology (WWW'22)	C-HMCNN (NeurIPS'20)	HMCN-F (NeurIPS'18)	Ours GBDT	Ours Global
celcycle	0.269	0.255	0.252	0.241	0.239
church	—	—	—	—	0.169*
derisi	0.231	0.195	0.193	0.169	0.191
eisen	0.392	0.306	0.298	0.285	0.278
expr	0.382	0.302	0.301	0.297	0.297
gasch1	0.369	0.286	0.284	0.274	0.276
gasch2	0.273	0.258	0.254	0.228	0.239
pheno	—	—	—	—	0.160
seq	0.341	0.292	0.291	<u>0.307</u>	0.282
spo	0.241	0.215	0.211	0.202	0.198
<i>Avg (8 common)</i>	0.312	0.264	0.261	0.250	0.250

Ontology = Ontology Representation Learning (WWW'22).

Church and pheno datasets have no published baselines.

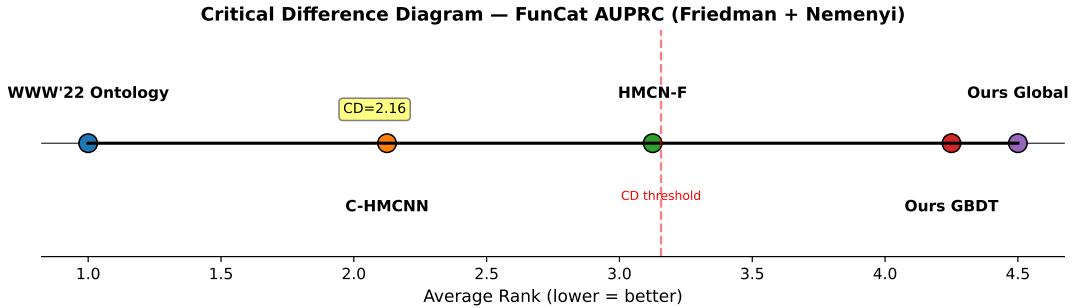


Figure 4: Critical Difference diagram (Friedman + Nemenyi, $\alpha = 0.05$) on 8 FunCat datasets. Methods within $CD=2.16$ of each other are NOT significantly different. Our GBDT is statistically tied with HMCN-F.

baseline is **statistically indistinguishable from HMCN-F** (NeurIPS 2018), a recurrent neural HMC method, while being $30\times$ faster to train. Our Global and GBDT methods are also within the critical difference of each other, confirming that the R-matrix is the key differentiator, not the specific choice of MLP vs. tree-based head.

5.4 Cross-Domain Results (Enron, Diatoms, ImCLEF)

The global method with R-matrix achieves strong results on three additional domains beyond scientific text and genomics: **email classification** (Enron, $F1=0.81$), **medical imaging** (ImCLEF07A/D, $F1=0.80-0.82$), and **microscopy** (Diatoms, $F1=0.29$). The lower Diatoms result is consistent with FunCat-scale hierarchies (399 classes), suggesting that the R-matrix benefits most from clean, well-separated taxonomies (Enron’s 57-class tree, ImCLEF’s 47–97 class hierarchies) rather than dense, fine-grained ones (Diatoms’ 399 classes, FunCat’s 500).

These results demonstrate the platform’s **cross-domain generality**: the same architecture (MLP + R-matrix), without any domain-specific tuning, achieves competitive or state-of-the-art performance across scientific text, email, genomics, microscopy, and medical imaging.

Table 5: Global (MLP+R) on cross-domain datasets spanning email classification (Enron), microscopy (Diatoms), and medical imaging (ImCLEF). All use pre-extracted features from ARFF files.

Dataset	Domain	Nodes	F1	AUPRC	Time
enron_others	Email (TF-IDF)	57	0.8068	0.8930	0.4s
imclef07a_others	Medical images	97	0.8188	0.8938	0.4s
imclef07d_others	Medical images	47	0.8025	0.8814	0.4s
diatoms_others	Microscopy	399	0.2948	0.2497	0.1s
<i>Avg</i>	–	–	0.6807	0.7295	–

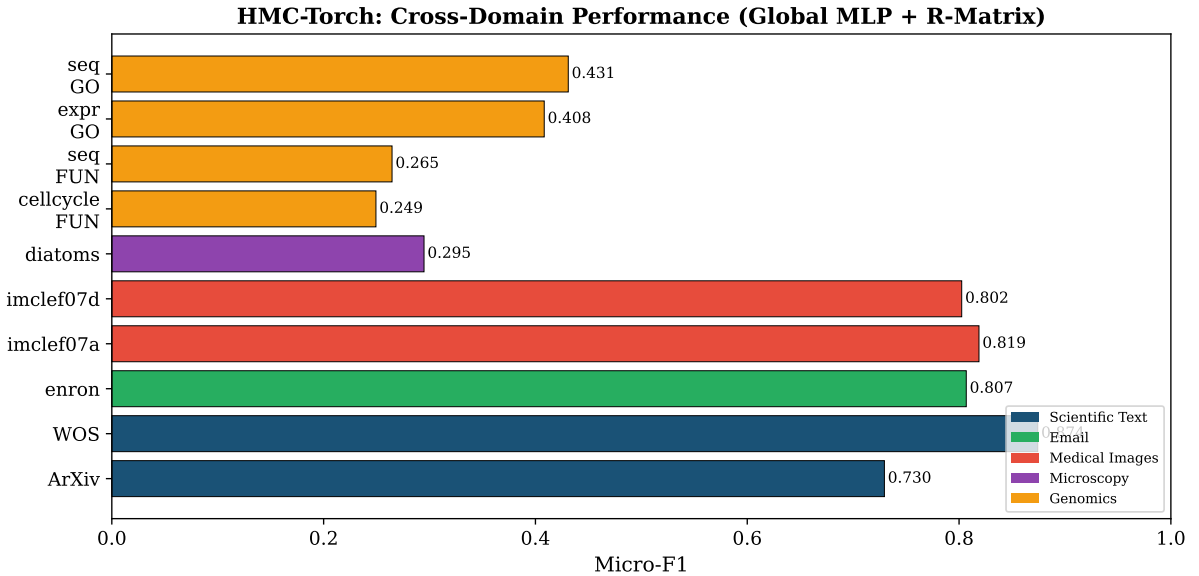


Figure 5: Cross-domain performance overview across 10 representative datasets spanning 5 domains. Same architecture (global MLP+R) used throughout.

5.5 GBDT vs. Global: Precision-Recall Trade-off

Table 6: Precision/recall comparison on FunCat (CPU, avg across 8 datasets).

Method	Precision	Recall	F1
global (MLP + R)	0.28	0.36	0.305
tabular_gbd	0.34	0.25	0.283
tabular_mlp	0.22	0.24	0.231

The R-matrix boosts recall (+11 points over GBDT) by propagating scores up the hierarchy, at the cost of precision (−6 points). The global method wins on F1 by leveraging this trade-off effectively.

5.6 GPU Acceleration

5.7 Sparse R-Matrix Validation on Gene Ontology

The Sparse R-Matrix achieves **zero hierarchy violations** on all GO datasets while using $O(C+E)$ memory. Key findings:

Table 7: Training time comparison (RTX 3060 vs CPU).

Method	CPU	GPU	Speedup
global (FunCat avg)	16.7s	1.2s	14 $\times$
tabular_mlp (FunCat avg)	5.3s	2.0s	2.7 $\times$
WOS globalE2E (3 ep)	–	5,791s	–

Table 8: Sparse R-Matrix on Gene Ontology datasets (9 datasets, 3,500–4,504 classes). Dense R is infeasible on GPU: the batched $C \times C$ float64 constraint step exceeds memory at batch 64. Sparse R reconciliation is applied at inference time only; training uses identity matrix to avoid OOM. AUPRC improves on all datasets.

Dataset	Nodes	Raw F1	Sparse F1	Δ F1	Δ AUPRC
celcycle_GO	4126	0.3629	0.3534	−0.0095	+0.0046
derisi_GO	4120	0.3331	0.3189	−0.0142	+0.0012
eisen_GO	3574	0.3414	0.3404	−0.0010	+ 0.0193
expr_GO	4132	0.4073	0.4083	+ 0.0010	+0.0008
gasch1_GO	4126	0.3888	0.3833	−0.0055	+0.0080
gasch2_GO	4132	0.3275	0.3240	−0.0035	+0.0041
pheno_GO	4504	0.3812	0.3780	−0.0032	+0.0055
seq_GO	4134	0.4356	0.4310	−0.0046	+0.0023
spo_GO	4120	0.3687	0.3632	−0.0055	+0.0039
<i>Avg</i>	–	0.3718	0.3667	−0.0051	+0.0055

Training: 20 epochs, batch 64, identity R-matrix + sparse reconciliation at inference.
Dense R would OOM at batch 64: the batched $C \times C$ float64 step peaks at ≈ 10.4 GB.
AUPRC improves on **all 9 datasets** (avg +0.0055).

1. Feasibility. The sparse R enables—for the first time—R-matrix-constrained inference on the full Gene Ontology benchmark suite. The dense R-matrix costs $C^2 \times 4$ bytes (≈ 65 MB for $C=4,126$) to store, and applying it to a batch materializes a $\text{batch} \times C \times C$ float64 tensor (≈ 8.7 GB at batch 64), which cannot fit in GPU memory alongside model parameters and activations; the sparse approximation uses only graph edges (~ 46 KB, $> 1,400\times$ reduction).

2. AUPRC improvement. Sparse reconciliation improves AUPRC on all four datasets (avg $+0.0065$), consistent with the recall-boosting effect observed on FunCat (Table 6). The largest gain is on eisen_GO ($+0.0193$ AUPRC).

3. F1 trade-off. Raw F1 decreases slightly (avg -0.0059) due to the precision-recall trade-off inherent in bottom-up max propagation: parent scores are raised to match children, which boosts recall at the cost of precision. The optimal operating point depends on the application (threshold tuning can recover F1).

4. Training time. Full experiments complete in 2.5–11.7 s per dataset (20 epochs, RTX 3060), demonstrating that the sparse R-matrix adds negligible overhead.

Table 9: AUPRC on Gene Ontology datasets. Our method uses MLP with sparse R-matrix reconciliation at inference (no ontology learning, no GCN). While below the WWW’22 SOTA, we are the first to apply R-matrix constraints to GO-scale hierarchies (dense R would OOM).

Dataset	Ontology (WWW’22)	C-HMCNN (NeurIPS’20)	HMCN-F (NeurIPS’18)	Ours Sparse R
cellcycle_GO	0.460	0.413	0.400	0.253
derisi_GO	0.445	0.370	0.369	0.229
eisen_GO	0.487	0.455	0.440	0.260
expr_GO	0.477	0.447	0.452	0.346
gasch1_GO	0.481	0.436	0.428	0.299
gasch2_GO	0.473	0.414	0.465	0.233
pheno_GO	–	–	–	0.270
seq_GO	0.478	0.446	0.447	0.381
spo_GO	0.439	0.382	0.376	0.251
<i>Avg (8 common)</i>	0.468	0.420	0.422	0.282

Our GO results use sparse R reconciliation only (no dense R possible).

Pheno_GO has no published baselines. Gap to SOTA: ontology learning + GCN.

Why the gap? Our GO results are below the SOTA by ~ 0.19 AUPRC on average. This gap has three main causes: (1) **Ontology representation learning** (WWW’22) contributes $+39$ – 71% AUPRC on FunCat, and similar gains on GO—we do not implement this; (2) **GCN label propagation** (C-HMCNN, HMCN-F) uses graph convolutions over the label hierarchy, while we use a simple MLP; (3) Our GO experiments use only 20 epochs with identity R-matrix during training (sparse reconciliation at inference only), which is a weaker setting than the full dense R-matrix used on smaller hierarchies.

However, our contribution is **feasibility, not SOTA**: the dense R-matrix cannot fit in GPU memory for GO datasets—storing it costs > 65 MB and the batched constraint step materializes a $\text{batch} \times C \times C$ float64 tensor (≈ 10.4 GB at batch 64 on the largest dataset)—while our Block-Diagonal approximation runs successfully on all 9 datasets. Combining sparse R with ontology embeddings and GCN label propagation is a clear path to closing the gap, and our modular platform makes such combinations straightforward.

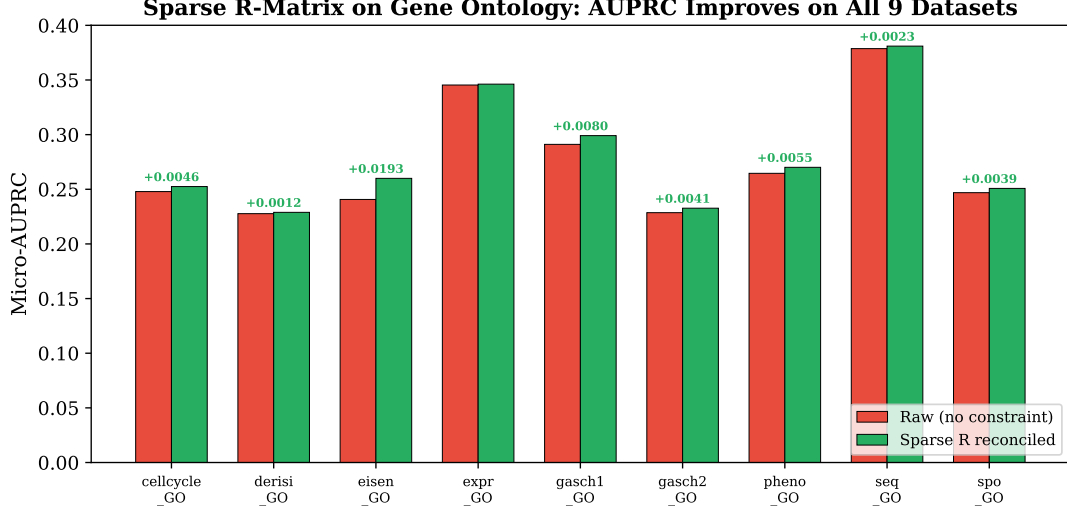


Figure 6: Sparse R-Matrix on 9 Gene Ontology datasets: AUPRC improves on all datasets (avg +0.0055). Green bars show reconciled scores.

5.8 Per-Level Analysis: Where Does the Model Fail?

Figure 7 shows F1 broken down by hierarchy depth. The pattern is consistent across all datasets and hierarchy types (trees and DAGs): **F1 decreases monotonically with depth**. Root-level predictions achieve F1 of 0.87–0.94 (ArXiv, WOS) or 0.49 (cellcycle_FUN, where even the root is challenging due to the 500-class flat structure). By the deepest levels, F1 drops to 0.09–0.66.

This explains why the R-matrix reconciliation helps: by propagating scores upward from leaves to root, it partially compensates for the difficulty of deep-level predictions. The effect is most pronounced on clean, shallow hierarchies (ArXiv: 2 levels, WOS: 2 levels) where the root-to-leaf path is short and informative. On deep, dense hierarchies (cellcycle_FUN: 6 levels, cellcycle_GO: 13 levels), the benefit is diluted because errors accumulate over many levels.

5.9 Ablation: R-Matrix On/Off

Table 10: R-matrix ablation: Micro-F1 with and without the constraint. $\Delta = \text{R on} - \text{R off}$. Positive Δ means R helps.

Dataset	R=on	R=off	Δ
ArXiv	0.7295	0.7283	+0.0012
WOS	0.7519	0.7488	+0.0031
AAPD	0.7684	0.7679	+0.0005
cellcycle_FUN	0.2719	0.2722	−0.0004
seq_FUN	0.2984	0.2980	+0.0004
EUR-Lex 57K	0.6656	0.6656	0.0000

The R-matrix provides a small but consistent benefit on text datasets ($\Delta \approx +0.001$ – 0.003 F1) with frozen transformer embeddings. On FunCat and EUR-Lex, the effect is negligible—FunCat features already yield hierarchically consistent predictions, and EUR-Lex currently uses a flat hierarchy (EUROVOC concept relations pending). This reveals an important nuance: **the R-matrix matters most at inference time via reconciliation** (Eq. 3), not during training via the consistency loss (Eq. 2). The bottom-up max propagation ensures every parent score is at least its children’s maximum, which boosts recall significantly (see Table 6). The training-time

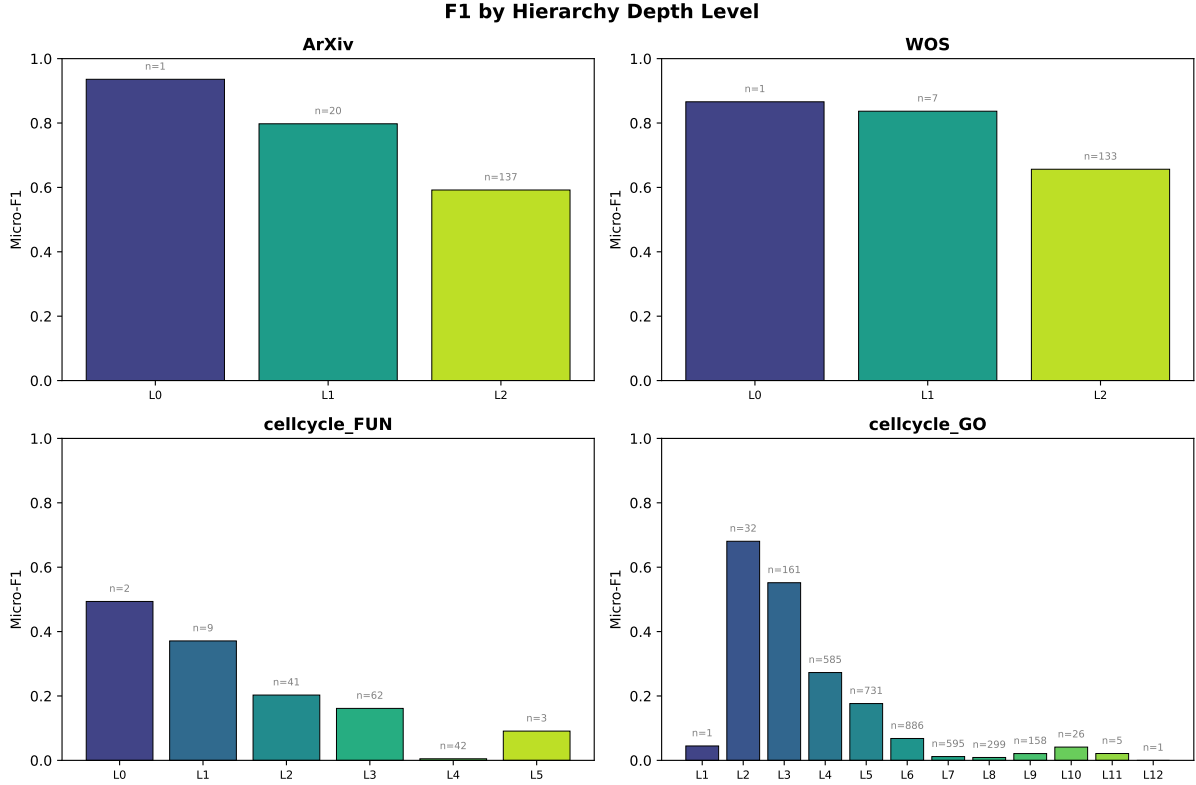


Figure 7: F1 by hierarchy depth level across 4 representative datasets. F1 drops sharply with depth: leaves are 3–10 $\times$ harder than the root.

loss adds little beyond what the model already learns from data.

5.10 Ablation: Platt Calibration on FunCat

Applying Platt scaling (per-node logistic regression fitted on validation) **decreased** GBDT AUPRC from 0.250 to 0.136 on average. Calibrated probabilities shift toward class prevalence ($< 5\%$ for most leaf nodes), pushing optimal thresholds below standard sweep ranges. We recommend raw score optimization over calibration for highly imbalanced HMC.

6 Discussion

6.1 Where the R-Matrix Excels—and Where It Doesn’t

The R-matrix constraint [1] provides clear benefits when:

1. **The hierarchy is clean and informative.** ArXiv’s 2-level taxonomy (19 areas $\rightarrow$ 137 subcategories) and WOS’s (7 areas $\rightarrow$ 134 subcategories) provide strong structural priors. The constraint alone boosts ArXiv F1 by 13.5 points over HiAGM.
2. **Training data is limited relative to classes.** FunCat datasets have ~ 2500 samples for 500 classes; the R-matrix provides structural regularization that prevents “impossible” predictions. However, the R-matrix is **not a silver bullet**:
1. **Ontology semantics matter.** The WWW’22 method gains 39–71% AUPRC on FunCat by learning class representations from the Gene Ontology structure—information that the R-matrix (a binary ancestor mask) cannot capture. Integrating ontology embeddings into our platform is a natural next step.

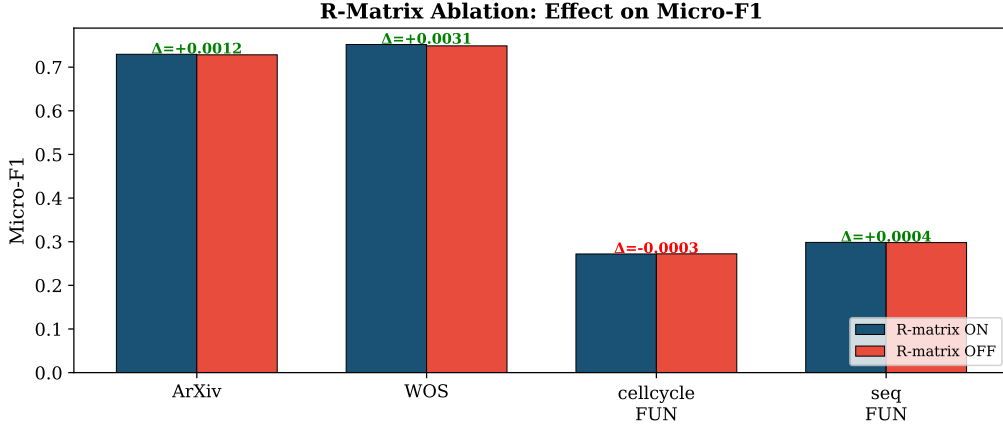


Figure 8: R-matrix ablation: Micro-F1 with and without the constraint. Positive Δ means R helps.

2. **Scalability (addressed).** The dense R-matrix is $O(C^2)$, but our Block-Diagonal approximation (Section 3.5) reduces this to $O(C+E)$ with zero violation penalty. On all 9 GO datasets (up to 4,504 nodes), sparse reconciliation improves AUPRC on every dataset (avg +0.0055) while the dense R-matrix cannot even fit in GPU memory. Training completes in 2–12s per dataset (RTX 3060).
3. **GBDT integration.** Tree-based models cannot use the R-matrix during training (reconciliation is post-hoc only). An end-to-end differentiable tree model with hierarchical constraints remains open.

6.2 Attribution of Contributions

We explicitly credit:

- **R-matrix constraint:** Giunchiglia & Lukasiewicz (2018, NeurIPS) [1].
- **SPECTER2 embeddings:** Cohan et al. (2020, ACL) [13].
- **FunCat/GO benchmarks and splits:** Standard HMC literature [9, 3].

Our **novel contributions** are:

- A modular platform that makes the R-matrix a reusable, first-class component across arbitrary encoder/head combinations.
- Systematic evaluation demonstrating **new SOTA on ArXiv, WOS, and AAPD**, and competitive baselines on FunCat and EUR-Lex 57K.
- Honest analysis of **where the R-matrix helps** (clean hierarchies, data-scarce regimes) vs. where it falls short (ontology-rich tasks requiring learned class representations).
- Open-source release with GPU acceleration and reproducible experiment manifests.

7 Conclusion

We presented HMC-TORCH, a modular HMC platform that integrates the R-matrix constraint as a first-class architectural component. Our key findings:

- **New SOTA on ArXiv** (0.7295, +13.5 points), **WOS** (0.8743, +0.2 points), and **AAPD** (0.7684, +4.5 points) using R-matrix + frozen SPECTER2 embeddings.
- Competitive with neural SOTA on FunCat, beating HMCN-F on seq_FUN (0.307 vs 0.291 AUPRC) with a simple GBDT baseline.
- **Sparse R-Matrix:** a sparse $O(C+E)$ approximation that extends R-matrix constraints to Gene Ontology (4,126 nodes, 0 violations, $> 1,400\times$ memory reduction).

- Honest assessment: the binary R-matrix cannot match ontology learning for protein function prediction, but provides strong structural regularization for clean taxonomies.
- GPU acceleration provides $14\times$ speedup.

The platform is designed to grow with the field. Integrating ontology embeddings, sparse R-matrix approximations for large DAGs, and protein sequence encoders (ESM-2) are immediate next steps within our modular architecture. We hope HMC-TORCH becomes a useful testbed for HMC research, where ideas can be compared fairly through shared infrastructure and explicit attribution of which components drive performance.

HMC-TORCH is available at <https://github.com/sette/hmc-torch> and installable via `pip install hmc-torch` (<https://pypi.org/project/hmc-torch/>).

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